Remote Image Bus (RIB) (S.O.O. 6212) – General data

1. General

The Remote Image Bus (RIB) module interfaces with external video inputs and the Modular Avionics Unit (MAU). The RIB module converts input analog video signals to a format required by Apex and the system displays. The RIB module is installed in the nose avionics bay.

2. Description

The RIB Video system consists of RIB switch/interface module, which is installed in the nose avionics bay (underside of left hand middle shelf). The interface module is capable of receiving one VGA signal and one composite video signal, which are converted to two RIB outputs. These outputs are connected to AGM 1 and AGM 2. The video signal output from the RIB module can be switched between VGA and composite using the input select pin on connector P477 of the RIB module. Ground selects the RGB input and open selects the composite input to be displayed.

The video input is displayed on the right 2/3rd of the Systems MFD; this display may be cycled between fuel/environment and video (default configuration) or charts, video and fuel/environment (upper MFD). No video cameras or video sources are included in S.O.O. 6212.

3. Adjustment/Test

A. Function Test RIB Module

1) Ensure that VIDEO (F7) (Left Bus) circuit breaker on the avionics circuit breaker panel is closed.

2) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

3) Wait several minutes for the aircraft electrical and avionics systems to initialize.

4) Set the system MFD to VIDEO by pressing the CHART/VIDEO button on the MF controller. If no video input source is connect to the RIB module, verify that a blank screen is displayed. If a video source is connected to the RIB module, verify the video feed is displayed.

5) Open circuit breaker VIDEO (F7) (Left Bus). Verify 'Unavailable' is displayed in the video window.

6) Close circuit breaker VIDEO (F7) (Left Bus). Verify a blank screen is displayed if there are not video inputs to the RIB module or the video feed is displayed in the video window if a video source is connected to the RIB module.

4. Removal/Installation

A. Removal of RIB Module

1) Ensure the aircraft electrical system is OFF.

2) Open VIDEO (F7) (Left Bus) circuit breaker on the avionics circuit breaker panel and fit safety clip.

3) Gain access to the forward avionics shelves through the forward baggage compartment.

4) Remove the middle HIRF cover.

5) Disconnect connectors connected to P1 (if installed), P2 and P3 of the RIB interface module.

6) While supporting the RIB interface module, remove the four mounting screws and remove unit from shelf.

Note Stow connectors as required.

B. Installation of RIB Module

1) Ensure the aircraft electrical system is OFF.

2) On the avionics circuit breaker panel ensure VIDEO (F7) (Left Bus) circuit breaker is open.

3) Set the RIB module in position and install the four mounting screws.

4) Reconnect electrical connectors.

5) Install the middle HIRF covers. Ensure the covers are mounted securely.

6) Remove the safety clip and close VIDEO (F7) (Left Bus) circuit breaker on the avionics circuit breaker panel.

7) Do a function test as per Para 3.A.

5. Troubleshooting

A. RIB Module

1) Ensure associated circuit breaker is closed.

2) Ensure connectors on the RIB module are fully seated and secured.

3) Visually inspect the connectors at the RIB module and MAU for damaged wiring or contacts.

4) If troubleshooting at the wire level is required, refer to the applicable wiring diagram in PSM 1-64-2W.

5) Verify input power voltage is 28 VDC.

6) If an input source is connected to the RIB module, verify the signal type is an RGB signal VGA (640x480) or composite signal (PAL or NTSC).